

Harith Al-Safi

Machine Learning Engineer

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📄 About me

A Software Engineer with 5 years of specializing in Machine Learning, React.js, Java Spring Boot and Cloud Technologies. Proven record in leading technical teams, and driving cost optimizations across **startup** to **enterprise**.

📁 Experience

Full Stack Engineer

📅 Sep 2024 - Ongoing (16+ Months)

BT Group

📍 Manchester, UK

- Delivered 5 **Kafka-based Spring Boot microservices** with MongoDB/PostgreSQL, Netflix Conductor, and Pulumi
- Mentored and examined 5+ apprentices on agile development with **Java 21** Spring Boot and **AWS Lambda**
- Achieved **30% AWS cost reduction** by automating infrastructure with TypeScript and optimizing ECS cluster
- Re-architecture Spring Boot APIs with JWT, Azure AD, and **role-based access** achieving **enterprise grade security**
- Led **React.js to Next.js** migration of 5K+ lines, adding Cypress testing and OpenAPI to TypeScript **code gen**
- Managed regional Graduate Network organizing and coordinating monthly tech talks for **30+ attendees**

Robotics Engineer

📅 Dec 2023 - Mar 2024 (4 Months)

Robot Fight League

📍 Leeds, UK

- Implemented a **control system** using Arduino and **C++** for the robot's movement, sensors and pneumatic system
- Tested obstacle detection and tracking using **OpenCV** and Python for the robot's **autonomous navigation** system
- Led **cross-functional** team of 5 engineers to 1st place finish, coordinating **hardware-software** integration

Software Engineer (Placement Year)

📅 July 2022 - July 2023 (12 Months)

Johnson Controls

📍 London, UK

- Invented a **scripting language with a compiler** for fire detection, increasing development productivity by 40%
- Managed a team of 10 interns to create a **business plan** and ended up placing 2nd amongst the whole company
- Administered C/C++ projects while fixing bugs, adding features, refactoring and **migrating 90% of codebase to C#**
- Developed C# .NET libraries for fire detection simulations and data analysis improving operations by 32%
- Collaborated on an **analytics web dashboard with React.js** for predictive maintenance with dev ops and CI/CD
- Maintained Python automation scripts for automated builds of .NET applications and processing data with Azure

Machine Learning Engineer

📅 Dec 2021 - May 2022 (6 Months)

Paperound - Freelance

📍 Leeds, UK

- Developed signal processing filters and modulators using SciPy Python and Fourier analysis for RF signals
- Built and trained a reinforcement neural network using TensorFlow, KerasRL for predicting market movements
- Applied data visualization using Python Matplotlib to present analytical insights to non-technical stakeholders
- Built production-ready Flask inference APIs packaged with Docker for low-latency model serving.
- Developed parallel processing system using **C++ OpenMP** to **accelerate polynomial regression** computations

Information Technology Administrator

📅 Jun 2020 - Jun 2021 (12 Months)

Mesopotamia Group

📍 Amman, Jordan

- Architected **scalable file server infrastructure** using Apache, Nextcloud, and Ubuntu in production environment
- Automated system administration workflows using **Python, Bash scripting**, and network security protocols
- Developed **high-availability load balancer with C/MPI** achieving 34% improved system uptime
- Built **predictive analytics pipeline using Python** for proactive cybersecurity threat detection
- Designed cloud networking solutions implementing TCP/IP, routing protocols, and infrastructure optimization

📖 Education

BEng (Honours) in Electronics and Computer

📅 Sept 2019 - July 2024

University of Leeds (UoL)

✅ First-Class Honours (4.0/4.0 GPA)

- Cloud and Parallel computing, Secure and Distributed systems, Cybersecurity, Software Engineering, Data Science
- Embedded systems, Microprocessors, FPGA, Circuit Analysis, Signal processing, Networking, Compiler design

Skills and Languages

Technical:

Python

C \ C++

C#

TypeScript

Java

Bash

Matplotlib

TensorFlow

Scikit-learn

Numpy

SciPy

Pandas

NLP

React.js

Next.js

Express.js

Redux

Tailwind CSS

Stripe

Auth0

Clerk

Spring Boot

Kafka

.NET

Docker

Jira

Git

MongoDB

PostgreSQL

Redis

SQLite

Jenkins

Firebase Firestore

AWS (RDS, Lambda, ECS, EC2, S3)

Azure (AD, MSAL, Serverless)

Pulumi

Soft:

Management

Leadership

Motivated

Deterministic

Communication

Problem solver

Languages:

Arabic (Full Professional)

English (Full Professional)

Spanish (Elementary)

Publications

Vega: LLM-driven Intelligent Chatbot Platform for Internet of Things Control and Development  2025

MPDI Sensors - 10.3390/s25123809 - 1st Author

- Developed a full stack **Chatbot** web application with TypeScript, React.js, Next.js, **TailwindCSS** and **Redis**
- Programmed language processing algorithms using **multi-modal large language models** to execute user messages
- Implemented a **Python Flask** REST Server with **SQLite** on the **Raspberry Pi** to communicate with the chatbot
- Engineered a **circuit** with sensors, camera, fans, LCD and buttons to execute user's commands from the chatbot

Projects

Green ML: Machine Learning for Green Bond Syndication  May 2024 - Aug 2024  Personal

- Engineered bond syndication **prediction model** using **XGBoost** and network analysis on 3K+ institutional holders
- Implemented **parallel processing** with Ray to extract network features from adjacency matrices with **70% speedup**
- Built machine learning pipeline achieving **0.95+ AUC-ROC** for predicting bond syndication collaboration patterns
- Processed **Bloomberg financial data** using **pandas**, **NetworkX** for multi-regional investment analysis

CV Studio  Oct 2024 - Ongoing  Personal - Beta on cvstudio.ai

- Built **full-stack** resume-improvement web app, powered by large language model, serving **hundreds of active users**.
- Implemented **live resume editing** with ShadcnUI and Typst **WebAssembly** PDF compilation for real-time render.
- Integrated **Clerk authentication** and **Stripe subscription** payments for secure user access and monetization
- Designed **Firestore real-time** database with Zustand state and Zod validation for seamless data syncing

MCPP: High Performance C++ Library for Numerical Computation  Apr 2021 - Oct 2022  Personal

- Created a **cross-platform math library** for C++ used for matrix algebra, statistics and machine learning.
- Engineered with **CI/CD** using **Jenkins**, unit tests, Docker build, **Cmake** and Doxygen documentation
- Improved performance by 40% using **parallel computing** with **OpenMP**, **OpenCL** and **CUDA**

Professional Courses

Data Analysis with Python  Oct 2021

- Develop Python code for cleaning and preparing data for analysis
- Perform exploratory **data analysis** and analytical techniques to datasets using Pandas, Numpy and Scipy
- Manipulate data using dataframes, understand data distribution, perform correlation and create data pipelines
- Build and evaluate **regression models** using machine learning scikit-learn for prediction and decision making

Databases and SQL for Data Science with Python  Sept 2021

- Analyze data within a relational database on the cloud using **IBM DB2**, SQL, Python
- Construct **SQL CRUD** operations and compose views, transactions, stored procedures and joins.

Python for Data Science, AI & Development  Aug 2021

- Demonstrate proficiency in using Python libraries such as Pandas, **Numpy**, and Beautiful Soup
- Access web data using APIs and **web scraping** from Python in Jupyter Notebooks.